
LECTURE NOTE

Lecture 4: Diffusion

AI508 - Computervision

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1 Evidence lower bound (ELBO)

Approximating ground truth latent variable encoder $p(z | x)$

We denote a variational distribution with parameter ϕ by $q_\phi(z | x)$

$$\log p(x) \geq \mathbb{E}_{q_\phi(x | x)} \left[\log \frac{p(x, z)}{q_\phi(z | x)} \right]$$

This lowerbound will make the encoder approximate the “ground truth”.

2 Markovian heriarchical VAEs

Generalize VAEs with multiple levels of latent variables. General hierarchical VAEs allow connection to all previous levels.

$$p(x, z_{1:T}) = p(z_T) p_\theta(x | z_1) \prod_{t=2}^T p_\theta(z_{t-1} | z_t)$$

$$q(z_{1:T}, x) = q_\theta(z_1 | x) \prod_{t=2}^T q_\theta(z_t | z_{t-1})$$

Markovian VAEs restrict to one-step transitions for both encoders and decoders

3 Variational diffusion models

A markovian hierarchical VAE with the following tree characteristics:

1. Latent dimensions = data dimension
2. The encoder is parameter—free and each step simply adds Gaussian noise
3. The amount of noise added varies such that x_T is standard Gaussian distributed

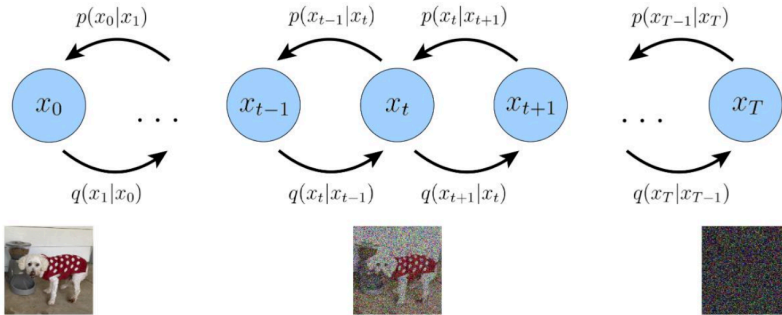


Figure 1: One can train a decoder to learn denoising on each step x_t or x_{t-1}

4 Conditional diffusion models

We rarely generate completely random images, so instead of $p(x)$ we are often interested in $p(x | y)$. Here y is some type of conditioning information.

Examples of conditioning information:

- Text embedding
- Low-resolution image
- Image with masked parts

So make encoder steps $p_\phi(x_{t-1} | x_t)$ Conditional

$$p_\phi(x_{t-1} | x_t, y)$$

5 Classifier guidance

At each step t , classify the noise latent x_t using a classifier. Classifier estimates to what degree x_t conforms to conditioning information y

$$\nabla \log p(x_t | y) = \underbrace{\nabla \log p(x_t)}_{\text{diffusion model}} + \gamma \underbrace{\nabla \log p(y | x_t)}_{\text{classifier}}$$

The classifier term adds a direction (*scaled by γ*) to the denoising step that increases the probability of the condition y , steering the sample toward y .

6 Classifier-free guidance

Combine a conditional and an unconditional diffusion model:

$$\nabla \log p(x_t | y) = \gamma \nabla \log p(x_t | y) + (1 - \gamma) \nabla \log p(x_t)$$

where λ control the importance attached to the conditioning

7 Image generation via conditional variation diffusion models (VDM)

Use text embeddings as conditional information. Most prominent is "Stable diffusion (SD)". Instead of embedding text alone, SD uses text→image embeddings (CLIP). Text embeddings are available via Unet, with cross-attention

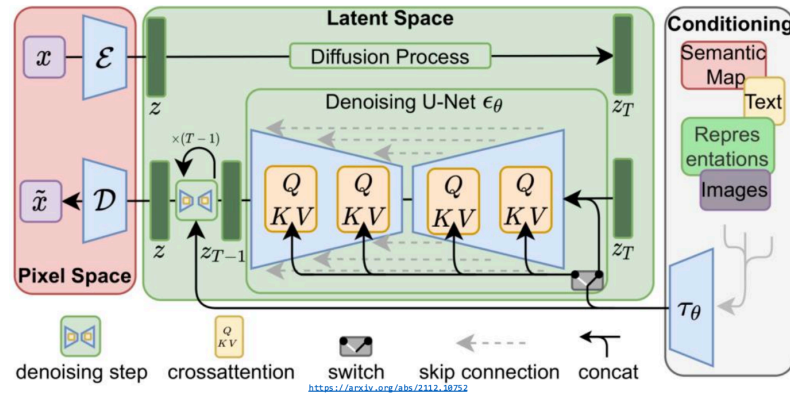


Figure 2: Example of a common CVDM
<https://arxiv.org/abs/2112.10752>

8 Inpainting via conditional variation diffusion models (VDM)

Use image as conditioning information. Parts of the images to be inpainted are obstructed.

Then at each step t of the denoising process, the conditioning information keeps the image grounded.



Figure 3: Inpainting examples
<https://doi.org/10.1145/3528233.3530757>